

Improving ER Efficiency Using Integer Linear Programming

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Abstract

Emergency Rooms (ERs) face the challenge of providing timely care while operating under staffing and budget limitations. When staffing does not match patient demand, wait times increase and operational efficiency decreases. This project applies an Integer Linear Programming (ILP) model to optimize the allocation of nurses and specialists across Morning, Afternoon, and Night shifts. Using a dataset of 5,000 ER patient visits from Kaggle, patient demand was calculated for each shift. The Afternoon shift had the highest demand, accounting for 64.54% of all visits. The ILP model maximized demand-weighted staffing while adhering to constraints related to minimum staffing levels, labor costs, staff availability, and a \$15,000 budget. The optimal solution allocated 39 staff members and fully consumed the budget. Sensitivity analysis showed that staffing improves with higher budgets until staff availability becomes the limiting factor. The results highlight the value of demand-based staffing and the importance of balancing cost, availability, and demand in ER operations.

Keywords: emergency room staffing, optimization, integer linear programming, healthcare operations, decision analytics.

Introduction

Emergency Rooms operate in high-pressure environments where staffing decisions directly affect patient care quality, wait times, and staff workload. ERs must ensure they have enough nurses and specialists to meet patient needs while staying within financial limits. This balance is difficult because patient demand varies across the day and staffing resources are fixed.

The purpose of this project is to develop an Integer Linear Programming model that identifies the best staffing plan across three shifts: Morning, Afternoon, and Night. The model

uses real patient demand patterns, cost assumptions, and operational constraints to recommend a staffing strategy that maximizes efficiency and remains realistic for hospital operations.

Project Background

The project uses a simulated ER Wait Time Dataset from Kaggle. The dataset contains 5,000 patient visits with fields such as Time of Day, Nurse-to-Patient Ratio, Specialist Availability, and Total Wait Time. These records were grouped into three operational shifts. The distribution of patient demand across the shifts was uneven. The Afternoon shift had the most demand, receiving 3,227 patients (64.54%). Morning received 1,243 patients (24.86%), and Night received 530 patients (10.60%).

This distribution of demand is important because it becomes the foundation of the objective function in the ILP model. The goal of the model is not just to assign enough staff to meet minimum requirements but to allocate staff proportionally to where demand is highest.

The central question of the project is: How can the ER assign nurses and specialists across shifts to better match patient demand while staying within budget and staffing limits?

Integer Linear Programming was selected because staffing decisions require whole numbers. A hospital can assign 5 or 6 nurses, but not 5.5 nurses.

Project Scope and Assumptions

Scope

The project focuses on two staff types: nurses and specialists and three daily shifts(Morning, Afternoon, Night). The model does not consider physicians, room availability,

triage severity, equipment, or non-labor resources. The emphasis is on labor allocation under realistic constraints.

Key Assumptions

Several assumptions were necessary to build the model:

- Historical patient counts represent future demand.
- Labor costs were assumed:
 - Nurse costs: \$300 (Morning), \$325 (Afternoon), \$375 (Night)
 - Specialist costs: \$500 (Morning), \$550 (Afternoon), \$650 (Night)
- Minimum staffing:
 - Nurses: Morning 4, Afternoon 8, Night 3
 - Specialists: Morning 1, Afternoon 2, Night 1
- Maximum available staff: 40 nurses and 10 specialists
- Overall staffing budget: \$15,000
- Specialists must not exceed nurses in any shift
- Afternoon requires at least one specialist for every three nurses

These assumptions create a realistic operational environment for the model.

Project Network and Activities

The project followed a structured analytic process. The steps included:

1. Collect and clean the dataset.
2. Group each patient visit into Morning, Afternoon, or Night based on time of day.
3. Count total patients per shift and calculate their share of total demand.

4. Define staffing requirements, labor costs, and budget.
5. Build the ILP model with decision variables, constraints, and an objective function.
6. Solve the model in R using the *lpSolve* package.
7. Analyze optimal staffing results.
8. Conduct sensitivity analysis on budget, demand, and nurse availability.
9. Develop recommendations.

The project network reflects a clear flow from data inputs to model outputs.

Linear Programming Model

Decision Variables

The model uses six decision variables:

- x_1 = Morning nurses
- x_2 = Afternoon nurses
- x_3 = Night nurses
- x_4 = Morning specialists
- x_5 = Afternoon specialists
- x_6 = Night specialists

Objective Function

The objective is to:

Maximize demand-weighted staffing.

This means the model gives more value to staff assigned to high-demand shifts. The weights are based on patient share:

- Morning: 0.2486
- Afternoon: 0.6454
- Night: 0.1060

Constraints

The model includes constraints covering:

- Minimum staffing requirements
- Staff availability
- Specialist-to-nurse balance rules
- Afternoon specialist ratio
- Total budget limit

These constraints ensure that the final staffing plan is both feasible and operationally realistic.

Analysis Results

The model returned a status of 0, meaning an optimal solution was found. The objective value was 20.6322.

Optimal Staffing Results

Shift	Assigned Nurses	Assigned Specialists	Total Staff
Morning	5	1	6
Afternoon	21	8	29
Night	3	1	4
Total	29	10	39

The Afternoon shift received the most staff because it had the highest patient demand.

Budget Usage

- Total staffing cost: \$15,000
- Budget remaining: \$0

The budget constraint was binding, meaning it fully limited the solution.

Shift	Patient Share	Staff Share
Morning	24.86%	15.38%
Afternoon	64.54%	74.36%
Night	10.60%	10.26%

The model appropriately prioritizes the Afternoon shift.

Sensitivity Analysis

Budget Sensitivity:

Budget levels from \$10,000 to \$20,000 were tested. Staffing increases with budget, but after \$18,375 in spending, staff availability becomes the bottleneck. At \$20,000, the model uses only \$18,375 because it cannot hire more than the available staff.

Demand Sensitivity:

Demand multipliers (0.80 to 1.50) were applied. Total staffing remained capped at 50 because staff availability was the limiting factor. Higher demand increased staffing in Morning and Night due to higher minimum requirements.

Nurse Availability Sensitivity:

When nurse availability was reduced to 20 nurses, total staffing dropped to 30. When availability increased to 40 nurses, total staffing rose to 50. This confirms that nurses are one of the most restrictive resources in ER operations.

Suggestions to Optimize

Prioritize Afternoon Staffing

Since this shift has the highest demand, it should receive the largest staffing allocation.

Use Flexible Staffing Options

Part-time, float pool, or on-call workers can reduce overstaffing in slower shifts while covering peak times.

Review the ER Budget

The base budget of \$15,000 was fully used. Increasing the budget may improve coverage but only if more staff are available.

Plan for Staff Availability Limits

Budget increases are only effective if more nurses and specialists can be recruited or scheduled.

Maintain Specialist Coverage

Specialists are heavily used in all scenarios. The ER should ensure strong recruitment and retention efforts.

Update the Model Regularly

As new patient data becomes available, the model should be recalibrated to reflect current demand.

Lessons Learned

Several important insights emerged from the analysis:

- Patient demand varies significantly across the day; staffing should reflect this.
- Demand-based staffing is more efficient than equal staffing across shifts.
- Budget and staff availability often limit performance more than demand.
- Nurse availability is a major bottleneck in ER operations.
- Optimization tools provide valuable guidance for data-driven staffing decisions.

Conclusion and Recommendations

This project developed and tested an Integer Linear Programming model to optimize ER staffing. The model used patient demand data, staffing requirements, cost assumptions, and operational constraints to produce a practical staffing plan. The optimal solution used all

available budgets and allocated most staff to the Afternoon shift, which had the highest patient demand.

Sensitivity analysis confirmed that staffing improves with more budget and more available nurses up to a limit. The findings support demand-based scheduling and emphasize the importance of flexible staffing and workforce planning.

The model provides a practical and data-driven approach to improving ER staffing efficiency and can be updated regularly as demand patterns evolve.

References

Rivalytics. (n.d.). *ER wait time dataset*. Kaggle.

<https://www.kaggle.com/datasets/rivalytics/er-wait-time>

Appendices

Appendix A: Dataset Summary by Shift

Shift	Total Patients	Average Patients per Nurse	Average Historical Specialist Availability	Demand Weight
Morning	1,243	3.243765	3.957361	0.2486
Afternoon	3,227	3.238612	3.857143	0.6454
Night	530	3.222642	3.790566	0.1060

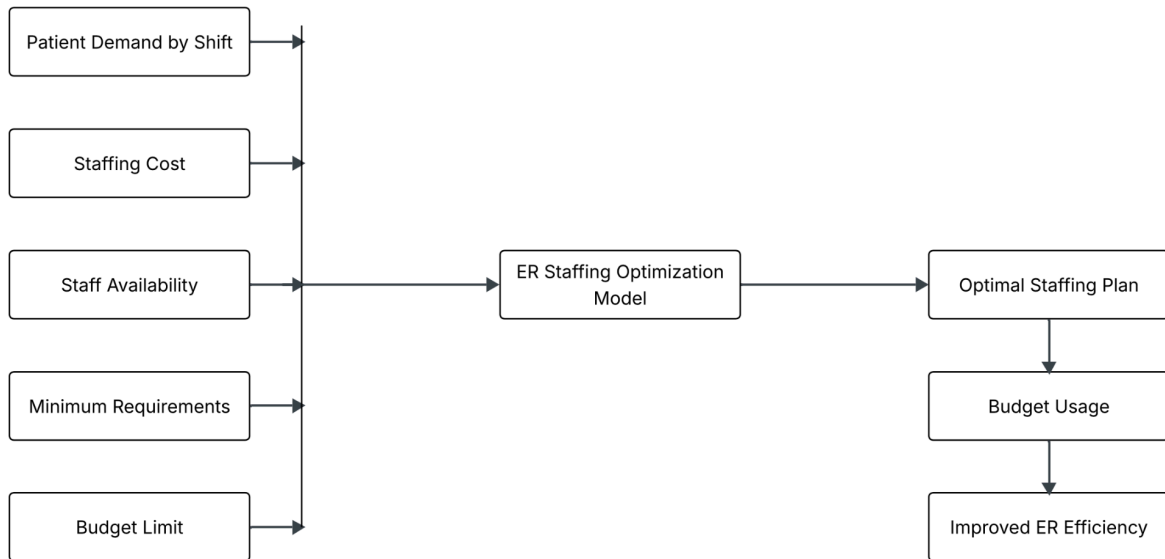
Appendix B: Base Optimal Staffing Solution

Shift	Required Nurses	Assigned Nurses	Required Specialists	Assigned Specialists	Total Staff
Morning	4	5	1	1	6
Afternoon	8	21	2	8	29
Night	3	3	1	1	4

Appendix C: Base Cost Summary

Shift	Nurse Cost	Specialist Cost	Total Staff	Total Shift Cost
Morning	\$300	\$500	6	\$2,000
Afternoon	\$325	\$550	29	\$11,225
Night	\$375	\$650	4	\$1,775
Total			39	\$15,000

Appendix D: Project Network Diagram (ER Staffing Optimization Model and Relationship Diagram)



Appendix E: R Console Output and Model Results

```

> # ER Staffing Integer Linear Programming Model

> # Objective: Maximize demand-weighted staffing under budget constraint

> library("lpSolve")

>

> # Read and prepare the ER wait time data

> er_data <- read.csv(file.choose())

>

> er_data$Shift <- ifelse(er_data$Time.of.Day %in% c("Early Morning", "Late Morning"),
"Morning",
+       ifelse(er_data$Time.of.Day %in% c("Afternoon", "Evening"), "Afternoon",
+       ifelse(er_data$Time.of.Day %in% c("Night", "Late Night"), "Night", NA)))

```

```
>

> er_data <- er_data[!is.na(er_data$Shift), ]

>

> # Aggregate shift information

> patient_counts <- aggregate(
+   Patient.ID ~ Shift,
+   data = er_data,
+   FUN = length
+ )

>

> names(patient_counts)[2] <- "Total_Patients"

>

> nurse_ratio_avg <- aggregate(
+   Nurse.to.Patient.Ratio ~ Shift,
+   data = er_data,
+   FUN = mean
+ )

>

> names(nurse_ratio_avg)[2] <- "Avg_Patients_Per_Nurse"

>

> specialist_avg <- aggregate(
+   Specialist.Availability ~ Shift,
+   data = er_data,
```

```

+ FUN = mean
+ )
>
> names(specialist_avg)[2] <- "Avg_Historical_Specialist_Availability"
>
> shift_summary <- merge(patient_counts, nurse_ratio_avg, by = "Shift")
> shift_summary <- merge(shift_summary, specialist_avg, by = "Shift")
>
> shift_order <- c("Morning", "Afternoon", "Night")
> shift_summary <- shift_summary[match(shift_order, shift_summary$Shift), ]
>
> print(shift_summary)

```

	Shift	Total_Patients	Avg_Patients_Per_Nurse
2	Morning	1243	3.243765
1	Afternoon	3227	3.238612
3	Night	530	3.222642

	Avg_Historical_Specialist_Availability
2	3.957361
1	3.857143
3	3.790566

```

>
> # Define staffing requirements
> required_nurses <- c(

```

```
+ Morning = 4,  
+ Afternoon = 8,  
+ Night = 3  
+ )  
>  
  
> required_specialists <- c(  
+ Morning = 1,  
+ Afternoon = 2,  
+ Night = 1  
+ )  
>  
  
> # Define shift labor costs  
  
> nurse_cost <- c(  
+ Morning = 300,  
+ Afternoon = 325,  
+ Night = 375  
+ )  
>  
  
> specialist_cost <- c(  
+ Morning = 500,  
+ Afternoon = 550,  
+ Night = 650  
+ )
```



```

>

> # Define available resources and budget

> total_nurses <- 40

> total_specialist <- 10

>

> available_budget <- 15000

>

> # Demand-weighted objective function

> # Calculate each shift's proportion of total patient demand.

> # Shifts with more patients receive larger objective weights.

> demand_weight <- shift_summary$Total_Patients / sum(shift_summary$Total_Patients)

> names(demand_weight) <- shift_summary$Shift

>

> print(demand_weight)

Morning Afternoon   Night

0.2486  0.6454  0.1060

>

> # Decision variables:

> # x1 = Morning nurses

> # x2 = Afternoon nurses

> # x3 = Night nurses

> # x4 = Morning specialists

> # x5 = Afternoon specialists

```

```

> # x6 = Night specialists

> #

> # Objective:

> # Maximize demand-weighted staffing.

> # This encourages the model to assign more staff to shifts

> # with higher patient demand.

>

> f.obj <- c(

+ demand_weight["Morning"],

+ demand_weight["Afternoon"],

+ demand_weight["Night"],

+ demand_weight["Morning"],

+ demand_weight["Afternoon"],

+ demand_weight["Night"]

+ )

>

> # integer linear program model

> f.con <- matrix(

+ c(

+ # Minimum nurse requirements

+ 1, 0, 0, 0, 0, 0,

+ 0, 1, 0, 0, 0, 0,

+ 0, 0, 1, 0, 0, 0,

```

```

+
+ # Minimum specialist requirements
+ 0, 0, 0, 1, 0, 0,
+ 0, 0, 0, 0, 1, 0,
+ 0, 0, 0, 0, 0, 1,
+
+ # Total available nurses and specialists
+ 1, 1, 1, 0, 0, 0,
+ 0, 0, 0, 1, 1, 1,
+
+ # Specialists must be less than or equal to nurses in each shift
+ # Morning:  $n1 - s1 \geq 0$ 
+ 1, 0, 0, -1, 0, 0,
+
+ # Afternoon:  $n2 - s2 \geq 0$ 
+ 0, 1, 0, 0, -1, 0,
+
+ # Night:  $n3 - s3 \geq 0$ 
+ 0, 0, 1, 0, 0, -1,
+
+ # Afternoon nurse-to-specialist ratio:
+ # At least 1 specialist per 3 nurses
+ #  $3s2 - n2 \geq 0$ 

```

```

+ 0, -1, 0, 0, 3, 0,
+
+ # Budget constraint:
+ # total staffing cost <= available_budget
+ nurse_cost["Morning"], nurse_cost["Afternoon"], nurse_cost["Night"],
+ specialist_cost["Morning"], specialist_cost["Afternoon"], specialist_cost["Night"]
+ ),
+ nrow = 13,
+ byrow = TRUE
+)
>
> f.dir <- c(
+ ">=", ">=", ">=",
+ ">=", ">=", ">=",
+ "<=", "<=",
+ ">=", ">=", ">=",
+ ">=",
+ "<="
+)
>
> f.rhs <- c(
+ required_nurses["Morning"],
+ required_nurses["Afternoon"],

```

```

+ required_nurses["Night"],
+ required_specialists["Morning"],
+ required_specialists["Afternoon"],
+ required_specialists["Night"],
+ total_nurses,
+ total_specialist,
+ 0,
+ 0,
+ 0,
+ 0,
+ available_budget
+ )
>

> er_staffing_sol <- lp(
+   direction = "max",
+   objective.in = f.obj,
+   const.mat = f.con,
+   const.dir = f.dir,
+   const.rhs = f.rhs,
+   all.int = TRUE
+ )
>

> # optimal staffing solution

```

```

> print(er_staffing_sol$status)

[1] 0

>

> # Status 0 means the model found an optimal feasible solution.

> # If status is not 0, the budget or staffing constraints may be infeasible.

> print(er_staffing_sol$objval)

[1] 20.6322

>

> opt_sol <- er_staffing_sol$solution

>

> names(opt_sol) <- c(
+ "Morning_Nurses", "Afternoon_Nurses", "Night_Nurses",
+ "Morning_Specialists", "Afternoon_Specialists", "Night_Specialists"
+ )

>

> print(opt_sol)

      Morning_Nurses  Afternoon_Nurses   Night_Nurses
              5           21              3

Morning_Specialists Afternoon_Specialists  Night_Specialists
              1              8              1

>

> # final solution table

> solution_table <- data.frame(

```

```

+ Shift = shift_order,
+ Total_Patients = shift_summary$Total_Patients,
+ Demand_Weight = demand_weight[shift_order],
+ Required_Nurses = required_nurses[shift_order],
+ Assigned_Nurses = opt_sol[1:3],
+ Required_Specialists = required_specialists[shift_order],
+ Assigned_Specialists = opt_sol[4:6],
+ Nurse_Cost_Per_Shift = nurse_cost[shift_order],
+ Specialist_Cost_Per_Shift = specialist_cost[shift_order]
+)
>
> solution_table$Total_Shift_Cost <-
+ solution_table$Assigned_Nurses * solution_table$Nurse_Cost_Per_Shift +
+ solution_table$Assigned_Specialists * solution_table$Specialist_Cost_Per_Shift
>
> solution_table$Total_Assigned_Staff <-
+ solution_table$Assigned_Nurses + solution_table$Assigned_Specialists
>
> print(solution_table)

```

	Shift	Total_Patients	Demand_Weight	Required_Nurses	Assigned_Nurses
Morning	Morning	1243	0.2486	4	5
Afternoon	Afternoon	3227	0.6454	8	21
Night	Night	530	0.1060	3	3

	Required_Specialists	Assigned_Specialists	Nurse_Cost_Per_Shift
Morning	1	1	300
Afternoon	2	8	325
Night	1	1	375

	Specialist_Cost_Per_Shift	Total_Shift_Cost	Total_Assigned_Staff
Morning	500	2000	6
Afternoon	550	11225	29
Night	650	1775	4

>

> # Display budget usage

> total_staffing_cost <- sum(solution_table\$Total_Shift_Cost)

>

> budget_table <- data.frame(

+ Available_Budget = available_budget,

+ Total_Staffing_Cost = total_staffing_cost,

+ Budget_Remaining = available_budget - total_staffing_cost

+)

>

> print(budget_table)

	Available_Budget	Total_Staffing_Cost	Budget_Remaining
1	15000	15000	0

>

> # Compare staffing with patient demand


```

> staffing_distribution <- data.frame(
+   Shift = shift_order,
+   Patient_Share = demand_weight[shift_order],
+   Staff_Assigned = solution_table$Total_Assigned_Staff,
+   Staff_Share = solution_table$Total_Assigned_Staff /
+     sum(solution_table$Total_Assigned_Staff)
+ )
>
> print(staffing_distribution)
      Shift Patient_Share Staff_Assigned Staff_Share
Morning  Morning    0.2486          6 0.1538462
Afternoon Afternoon  0.6454         29 0.7435897
Night    Night    0.1060          4 0.1025641
>
>
> # SCENARIO 1
> # Budget Sensitivity Analysis
> # This scenario tests how staffing changes when the available budget changes.
> # The objective remains the same:
> # maximize demand-weighted staffing.
> budget_levels <- c(10000, 12500, 15000, 17500, 20000)
>
> budget_scenario_results <- data.frame()

```

```

>
> for (b in budget_levels) {
+
+   available_budget <- b
+
+   f.rhs <- c(
+     required_nurses["Morning"],
+     required_nurses["Afternoon"],
+     required_nurses["Night"],
+     required_specialists["Morning"],
+     required_specialists["Afternoon"],
+     required_specialists["Night"],
+     total_nurses,
+     total_specialist,
+     0,
+     0,
+     0,
+     0,
+     available_budget
+   )
+
+   budget_sol <- lp(
+     direction = "max",

```

```

+ objective.in = f.obj,
+ const.mat = f.con,
+ const.dir = f.dir,
+ const.rhs = f.rhs,
+ all.int = TRUE
+ )
+
+ opt <- budget_sol$solution
+
+ total_cost <- sum(
+   opt[1:3] * nurse_cost[shift_order] +
+   opt[4:6] * specialist_cost[shift_order]
+ )
+
+ scenario_row <- data.frame(
+   Budget = b,
+   Status = budget_sol$status,
+   Objective_Value = budget_sol$objval,
+   Morning_Nurses = opt[1],
+   Afternoon_Nurses = opt[2],
+   Night_Nurses = opt[3],
+   Morning_Specialists = opt[4],
+   Afternoon_Specialists = opt[5],

```

```

+   Night_Specialists = opt[6],
+   Total_Nurses = sum(opt[1:3]),
+   Total_Specialists = sum(opt[4:6]),
+   Total_Staff = sum(opt),
+   Total_Cost = total_cost,
+   Budget_Remaining = b - total_cost
+ )
+
+ budget_scenario_results <- rbind(
+   budget_scenario_results,
+   scenario_row
+ )
+ }
+
>
> print(budget_scenario_results)

```

	Budget	Status	Objective_Value	Morning_Nurses	Afternoon_Nurses	Night_Nurses
1	10000	0	12.2420	5	12	3
2	12500	0	16.5112	4	17	3
3	15000	0	20.6322	5	21	3
4	17500	0	23.8114	10	24	3
5	20000	0	24.5572	13	24	3

	Morning_Specialists	Afternoon_Specialists	Night_Specialists	Total_Nurses
1	1	4	1	20

2	1	6	1	24
3	1	8	1	29
4	1	8	1	37
5	1	8	1	40

Total_Specialists Total_Staff Total_Cost Budget_Remaining

1	6	26	9875	125
2	8	32	12300	200
3	10	39	15000	0
4	10	47	17475	25
5	10	50	18375	1625

>

>

>

> # SCENARIO 2

> # Patient Demand Increase / Decrease

> # This scenario tests how staffing changes when patient demand changes.

> # Unlike changing only the objective weights, this changes the minimum

> # required staffing levels.

>

> demand_multipliers <- c(0.80, 1.00, 1.20, 1.50)

>

> demand_scenario_results <- data.frame()

>

```

> for (m in demand_multipliers) {
+
+   adjusted_required_nurses <- ceiling(required_nurses * m)
+   adjusted_required_specialists <- ceiling(required_specialists * m)
+
+   f.rhs.demand <- c(
+     adjusted_required_nurses["Morning"],
+     adjusted_required_nurses["Afternoon"],
+     adjusted_required_nurses["Night"],
+     adjusted_required_specialists["Morning"],
+     adjusted_required_specialists["Afternoon"],
+     adjusted_required_specialists["Night"],
+     total_nurses,
+     total_specialist,
+     0,
+     0,
+     0,
+     0,
+     available_budget
+   )
+
+   demand_sol <- lp(
+     direction = "max",

```

```

+ objective.in = f.obj,
+ const.mat = f.con,
+ const.dir = f.dir,
+ const.rhs = f.rhs.demand,
+ all.int = TRUE
+ )
+
+ opt <- demand_sol$solution
+
+ total_cost <- sum(
+   opt[1:3] * nurse_cost[shift_order] +
+   opt[4:6] * specialist_cost[shift_order]
+ )
+
+ scenario_row <- data.frame(
+   Demand_Multiplier = m,
+   Status = demand_sol$status,
+   Objective_Value = demand_sol$objval,
+   Morning_Nurses = opt[1],
+   Afternoon_Nurses = opt[2],
+   Night_Nurses = opt[3],
+   Morning_Specialists = opt[4],
+   Afternoon_Specialists = opt[5],

```

```

+   Night_Specialists = opt[6],
+   Total_Nurses = sum(opt[1:3]),
+   Total_Specialists = sum(opt[4:6]),
+   Total_Staff = sum(opt),
+   Total_Cost = total_cost,
+   Budget_Remaining = available_budget - total_cost
+ )
+
+ demand_scenario_results <- rbind(
+   demand_scenario_results,
+   scenario_row
+ )
+ }
>
> print(demand_scenario_results)

Demand_Multiplier Status Objective_Value Morning_Nurses Afternoon_Nurses
1      0.8      0      24.5572      13      24
2      1.0      0      24.5572      13      24
3      1.2      0      21.0976      18      18
4      1.5      0      20.9550      17      18

Night_Nurses Morning_Specialists Afternoon_Specialists Night_Specialists
1      3      1      8      1
2      3      1      8      1

```


3 4 2 6 2

4 5 2 6 2

Total_Nurses Total_Specialists Total_Staff Total_Cost Budget_Remaining

1 40 10 50 18375 1625

2 40 10 50 18375 1625

3 40 10 50 18350 1650

4 40 10 50 18425 1575

>

>

> # SCENARIO 3

> # Nurse Availability Shortage

> # This scenario tests how staffing changes when the total number of available nurses changes.

> # The objective remains the same:

> # maximize demand-weighted staffing.

> nurse_availability_levels <- c(20, 25, 30, 35, 40)

>

> nurse_availability_results <- data.frame()

>

> for (nurse_limit in nurse_availability_levels) {

+

+ f.rhs.nurse <- c(

+ required_nurses["Morning"],

+ required_nurses["Afternoon"],

```

+   required_nurses["Night"],
+   required_specialists["Morning"],
+   required_specialists["Afternoon"],
+   required_specialists["Night"],
+   nurse_limit,
+   total_specialist,
+   0,
+   0,
+   0,
+   0,
+   available_budget
+ )
+
+ nurse_sol <- lp(
+   direction = "max",
+   objective.in = f.obj,
+   const.mat = f.con,
+   const.dir = f.dir,
+   const.rhs = f.rhs.nurse,
+   all.int = TRUE
+ )
+
+ opt <- nurse_sol$solution

```

```

+
+ total_cost <- sum(
+   opt[1:3] * nurse_cost[shift_order] +
+   opt[4:6] * specialist_cost[shift_order]
+ )
+
+ scenario_row <- data.frame(
+   Nurse_Availability = nurse_limit,
+   Status = nurse_sol$status,
+   Objective_Value = nurse_sol$objval,
+   Morning_Nurses = opt[1],
+   Afternoon_Nurses = opt[2],
+   Night_Nurses = opt[3],
+   Morning_Specialists = opt[4],
+   Afternoon_Specialists = opt[5],
+   Night_Specialists = opt[6],
+   Total_Nurses = sum(opt[1:3]),
+   Total_Specialists = sum(opt[4:6]),
+   Total_Staff = sum(opt),
+   Total_Cost = total_cost,
+   Budget_Remaining = available_budget - total_cost
+ )
+

```

```

+ nurse_availability_results <- rbind(
+   nurse_availability_results,
+   scenario_row
+ )
+ }
>

```

```
> print(nurse_availability_results)
```

	Nurse_Availability	Status	Objective_Value	Morning_Nurses	Afternoon_Nurses
1	20	0	15.2204	4	13
2	25	0	18.4474	4	18
3	30	0	21.6744	4	23
4	35	0	23.3142	8	24
5	40	0	24.5572	13	24

	Night_Nurses	Morning_Specialists	Afternoon_Specialists	Night_Specialists
1	3	1	8	1
2	3	1	8	1
3	3	1	8	1
4	3	1	8	1
5	3	1	8	1

	Total_Nurses	Total_Specialists	Total_Staff	Total_Cost	Budget_Remaining
1	20	10	30	12100	7900
2	25	10	35	13725	6275
3	30	10	40	15350	4650

4	35	10	45	16875	3125
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5	40	10	50	18375	1625
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>